

General classification performance metrics

- They can be applied to **any kind of classification model**, not just logistic regression.
- These metrics are commonly used for evaluating the performance of models that deal with **categorical outcomes** (binary or multiclass).

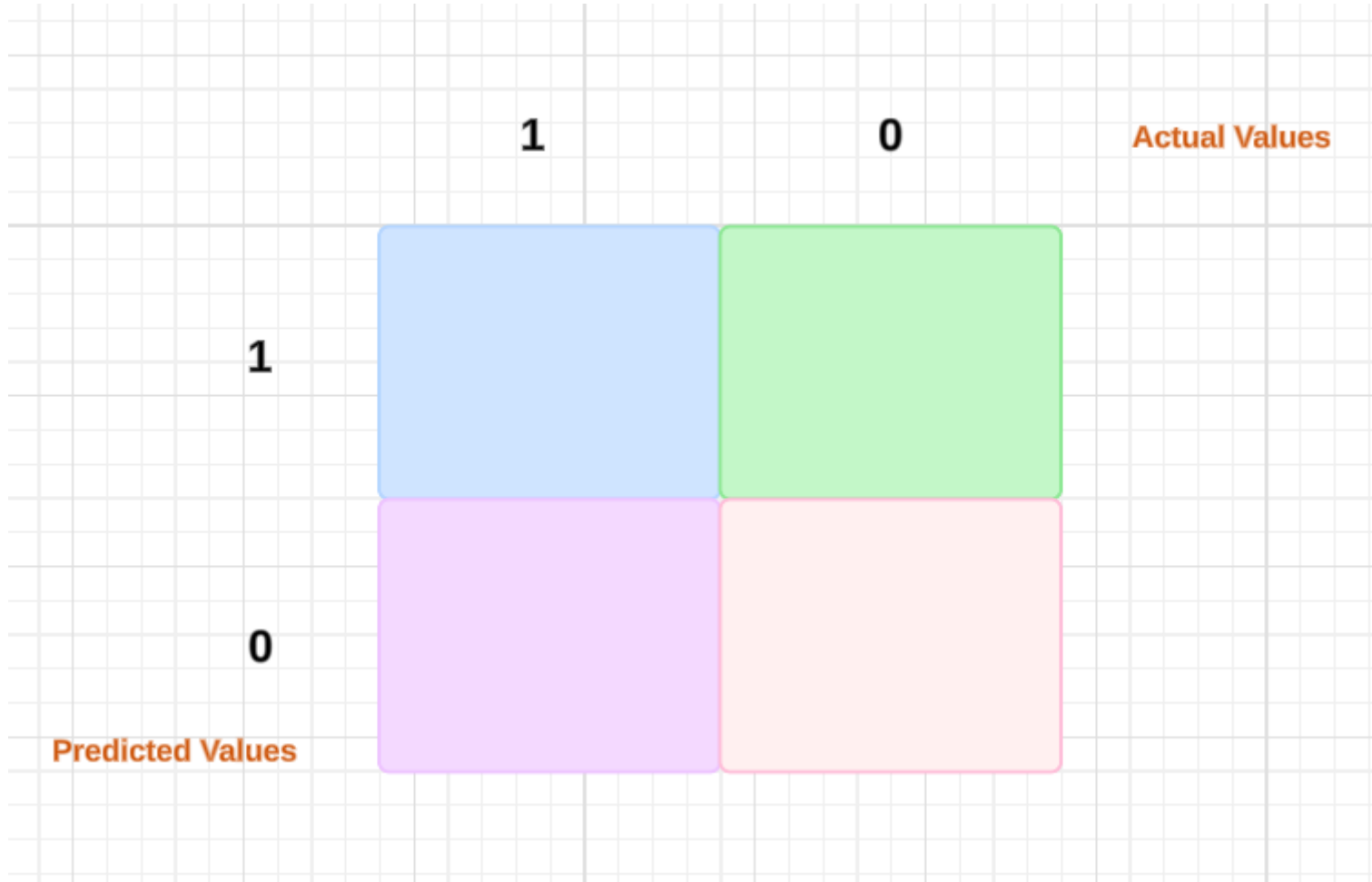
.....

- 1. Confusion Matrix**
- 2. Accuracy**
- 3. Precision**
- 4. Recall**
- 5. F-Beta Score**

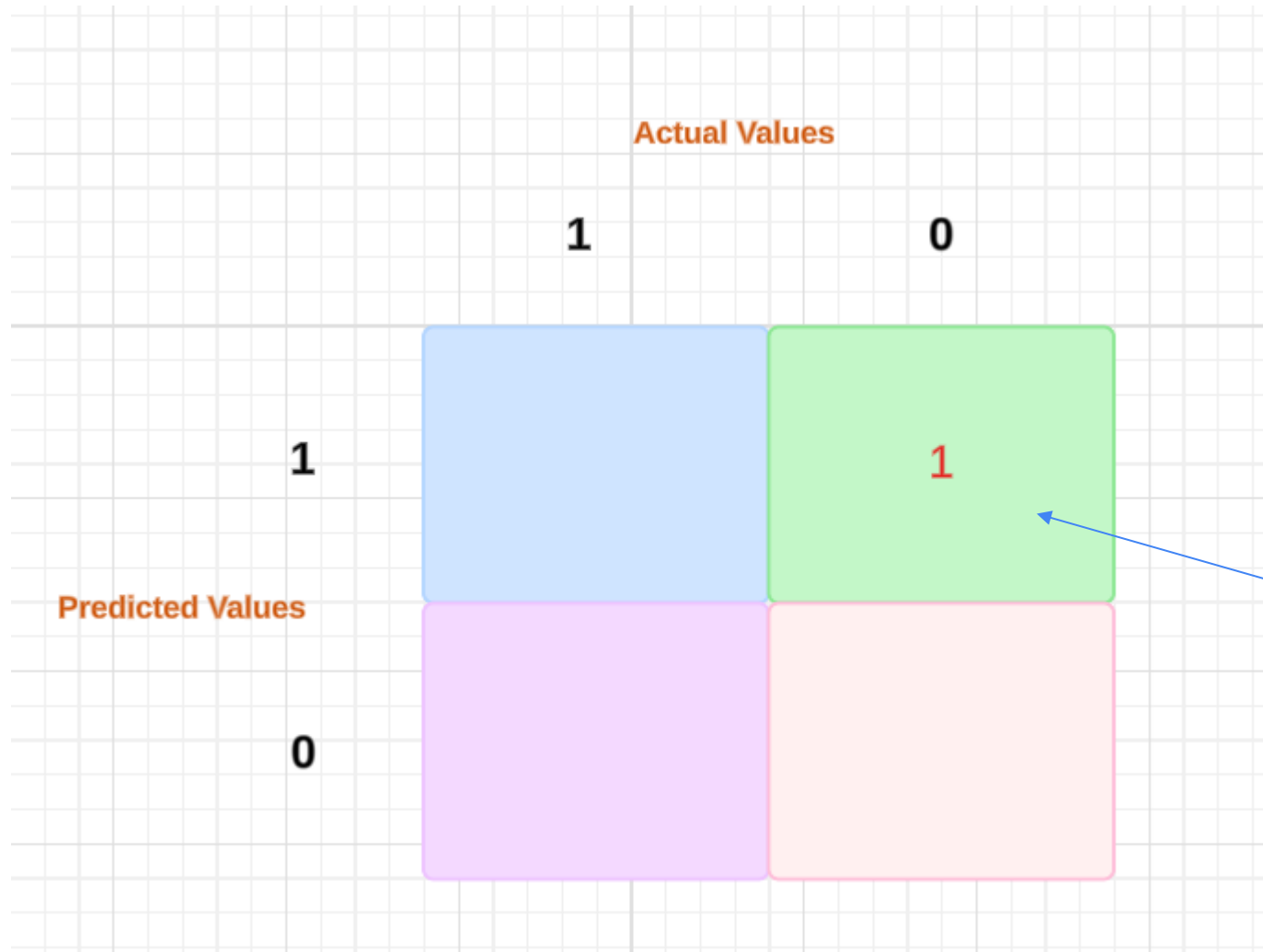
■ ■ ■ ■ ■ ■ ■ ■ ■ ■

X1	X2	Y	
A	Z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

Confusion Matrix

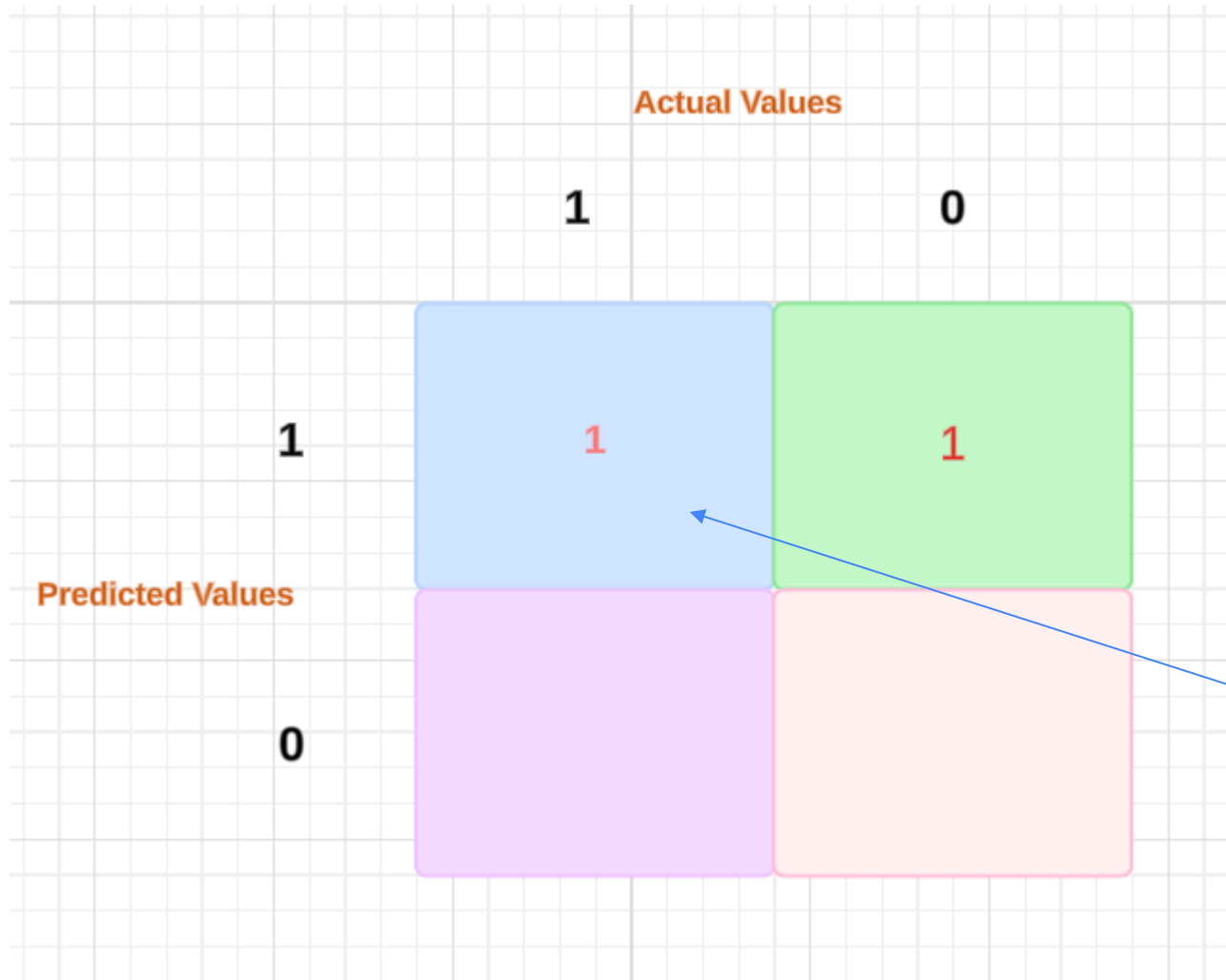


This is a 2 X 2 Matrix



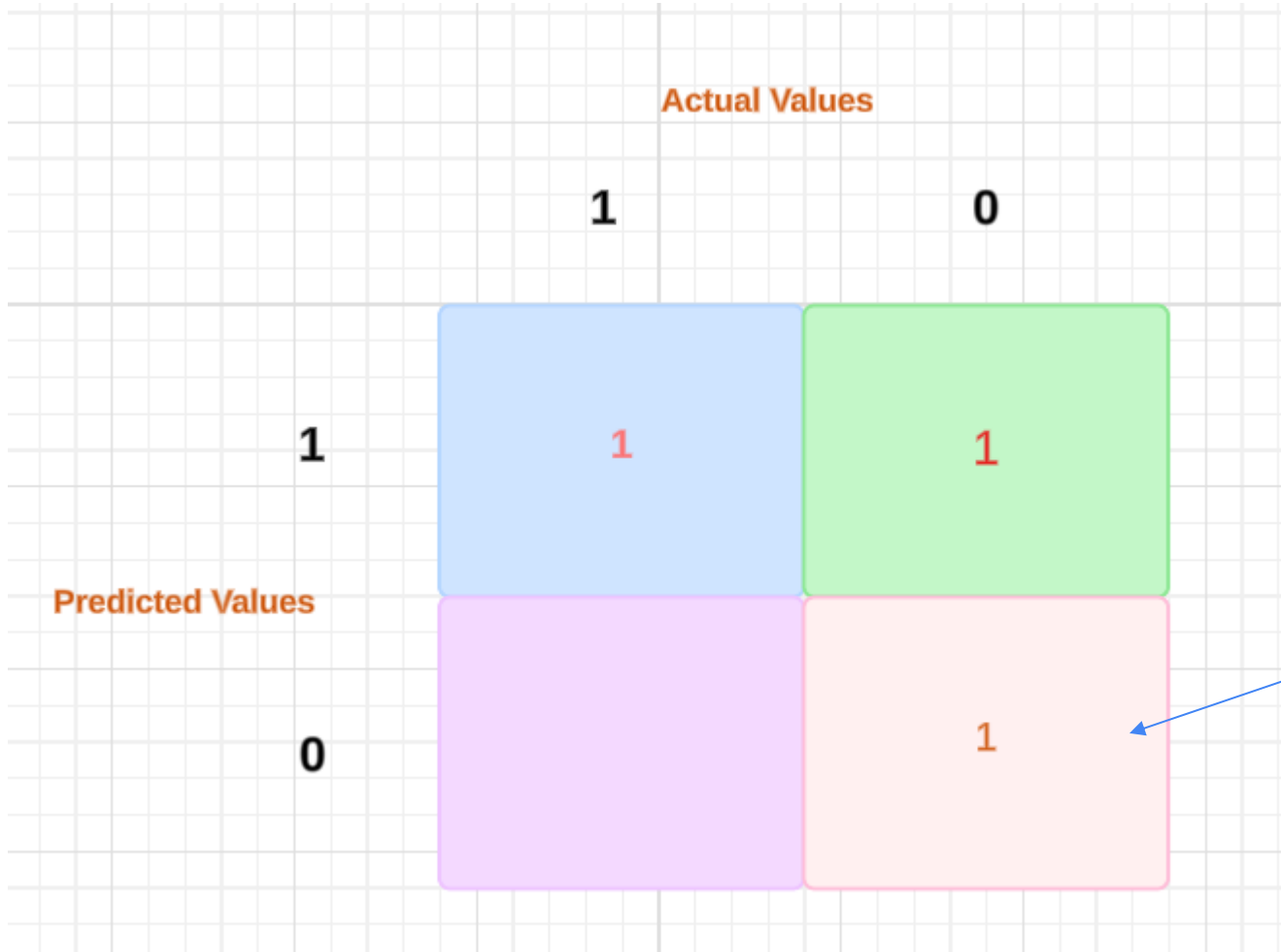
X1	X2	Y	$\hat{y}$
A	z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

My Actual output is **0** but our Model predicted as **1**



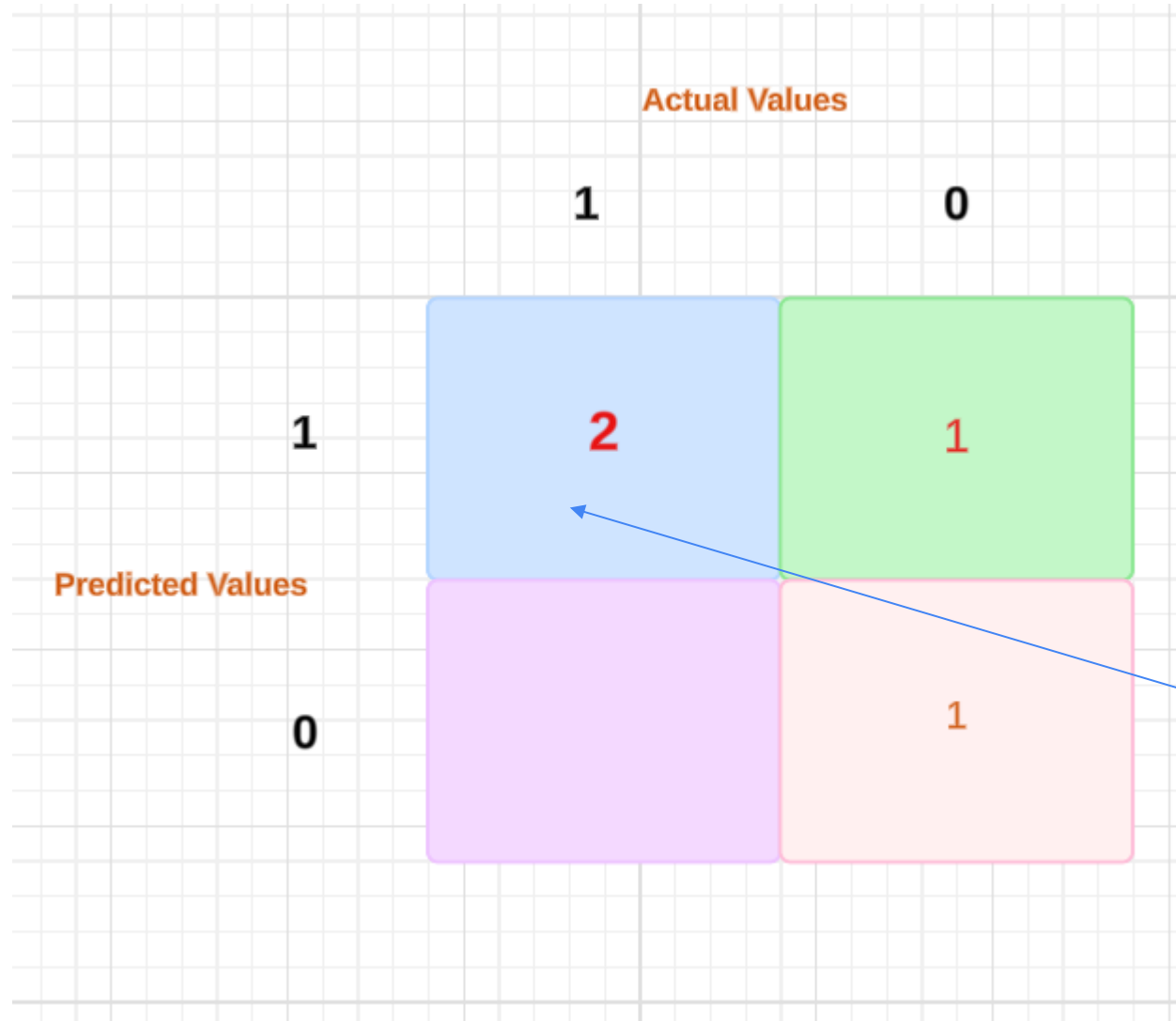
	X1	X2	Y	$\hat{y}$
A	Z		0	1
B	c		1	1
k	h		0	0
u	l		1	1
f	y		1	1
i	o		0	1
g	z		1	0

My Actual output is **1** our Model also predicted as **1**



X1	X2	Y	$\hat{y}$
A	Z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

My Actual output is **0** our
Model also predicted as **0**

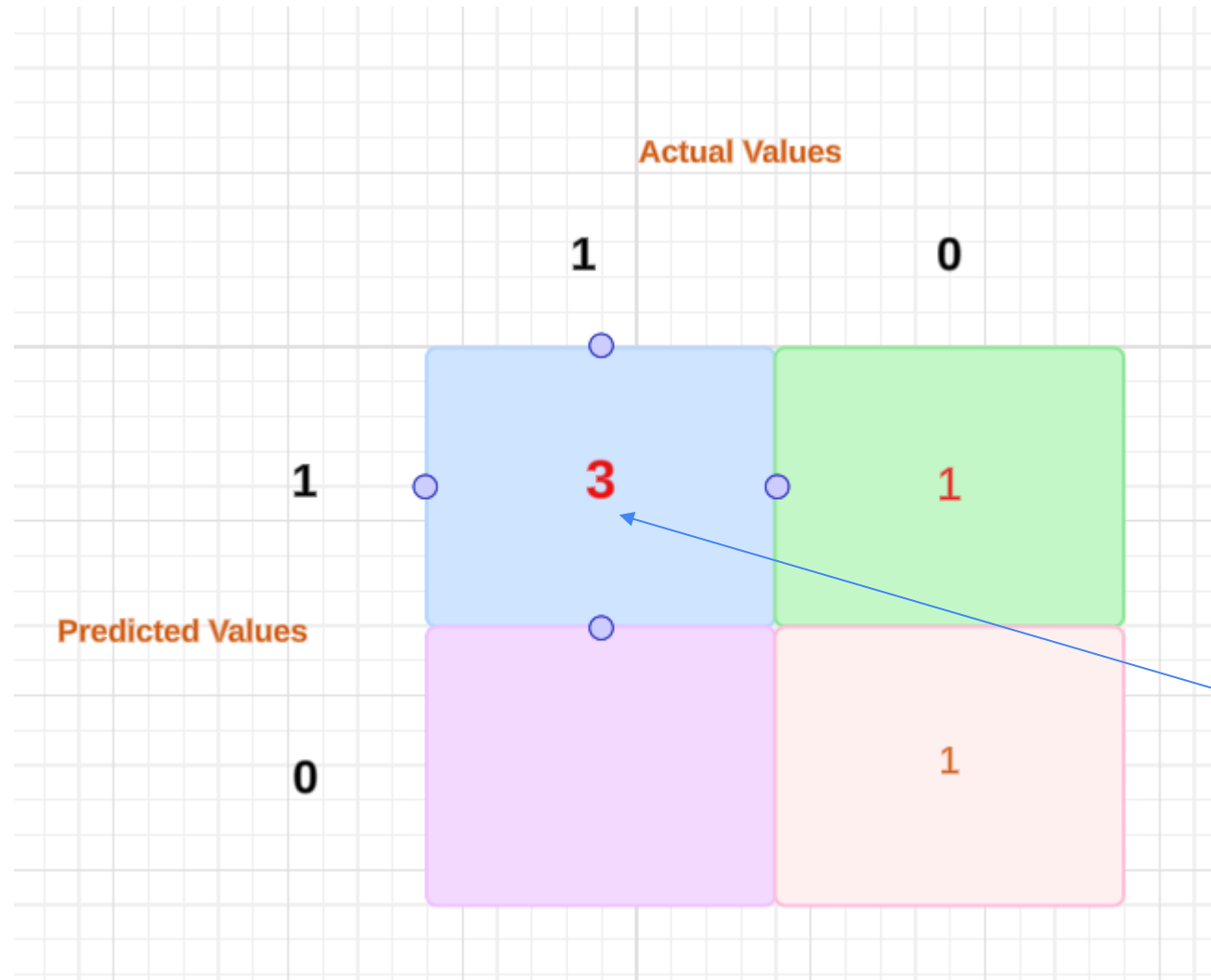


X1	X2	Y	$\hat{y}$
A	Z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

My Actual output is **1** our Model also predicted as **1**

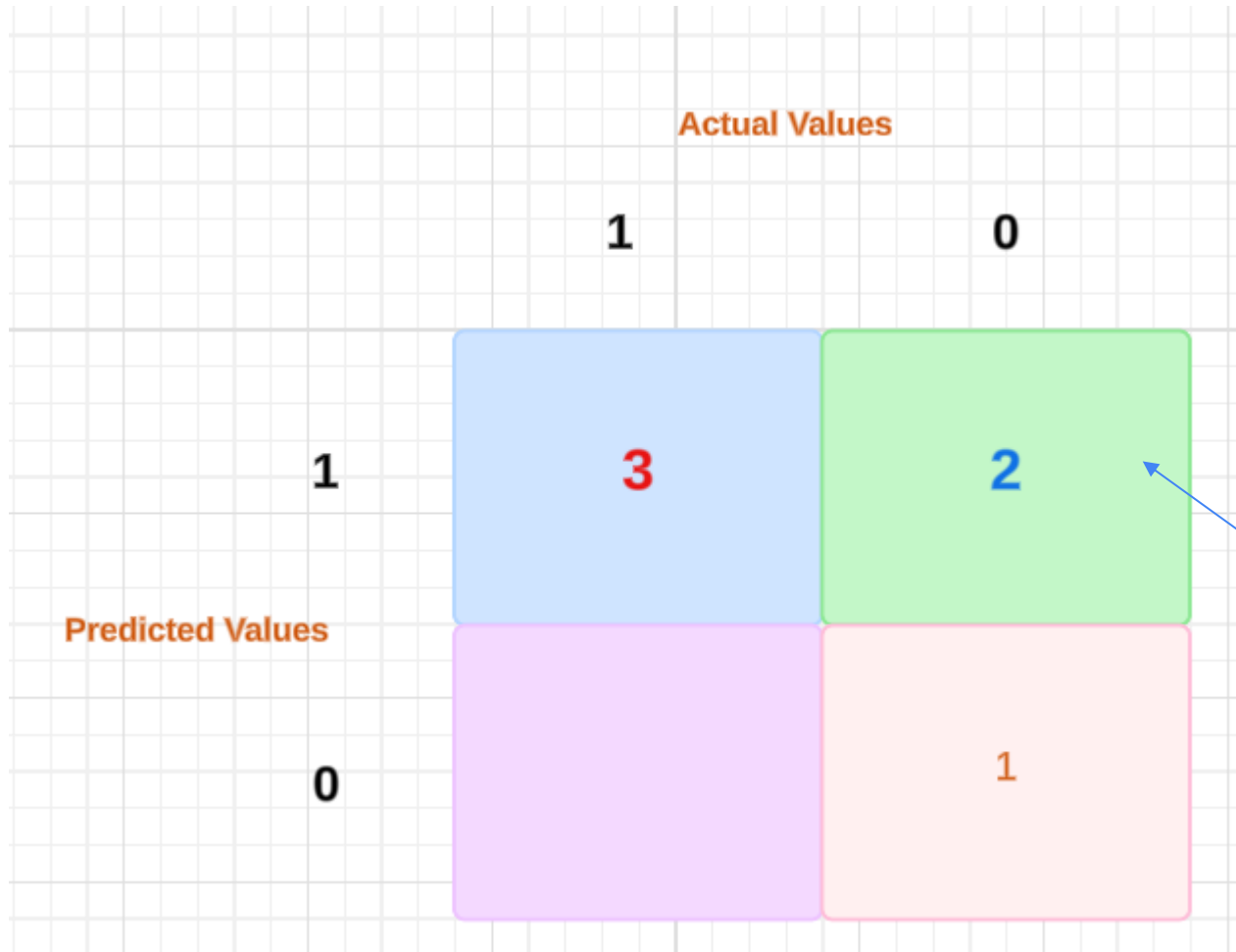
Now increment the value in the matrix

	X1	X2	Y	$\hat{y}$
A	Z		0	1
B	c		1	1
k	h		0	0
u	l		1	1
f	y		1	1
i	o		0	1
g	z		1	0



Again My Actual output is **1**
our Model also predicted as **1**

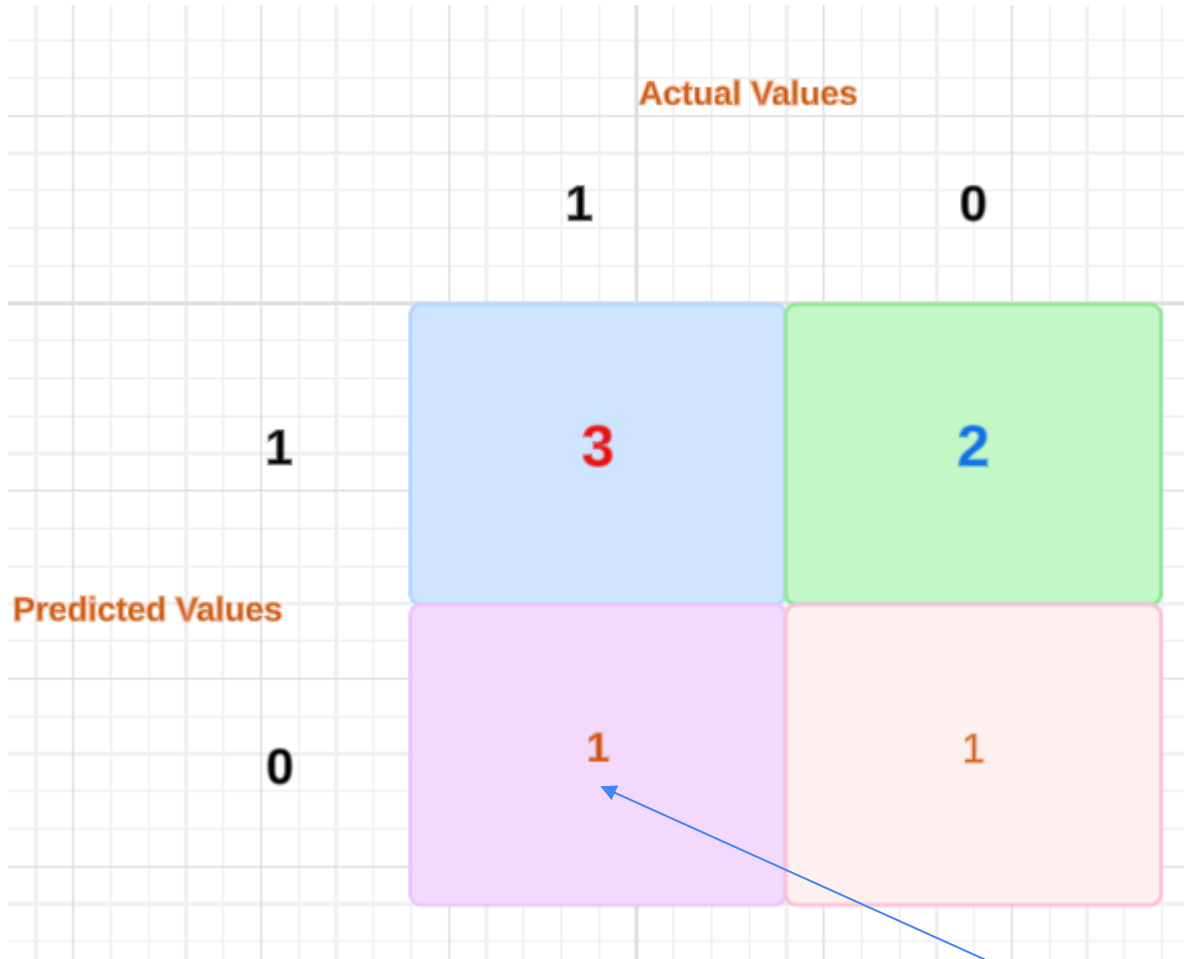
**Now increment the value in
the matrix**



X1	X2	Y	$\hat{y}$
A	Z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

My Actual output is **0**
our Model predicted as **1**

Now increment the
value in the matrix



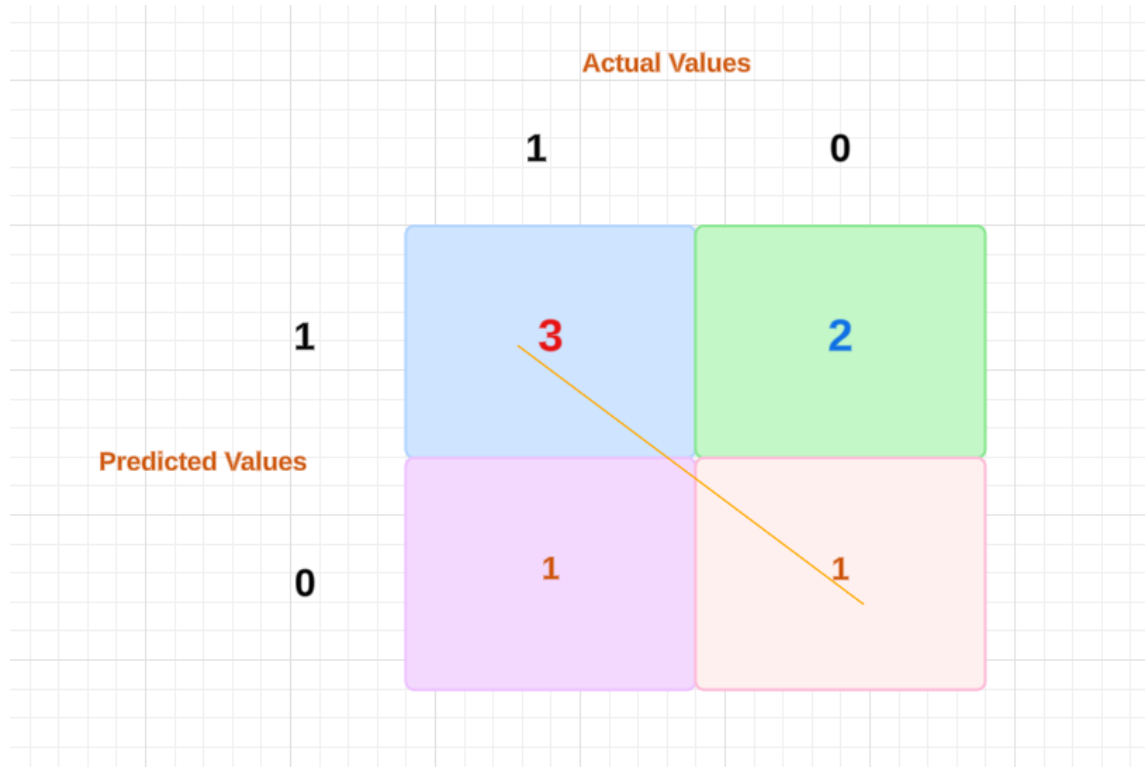
	X1	X2	Y	$\hat{y}$
A	Z		0	1
B	c		1	1
k	h		0	0
u	l		1	1
f	y		1	1
i	o		0	1
g	z		1	0

My Actual output is **1**
our Model predicted as
0

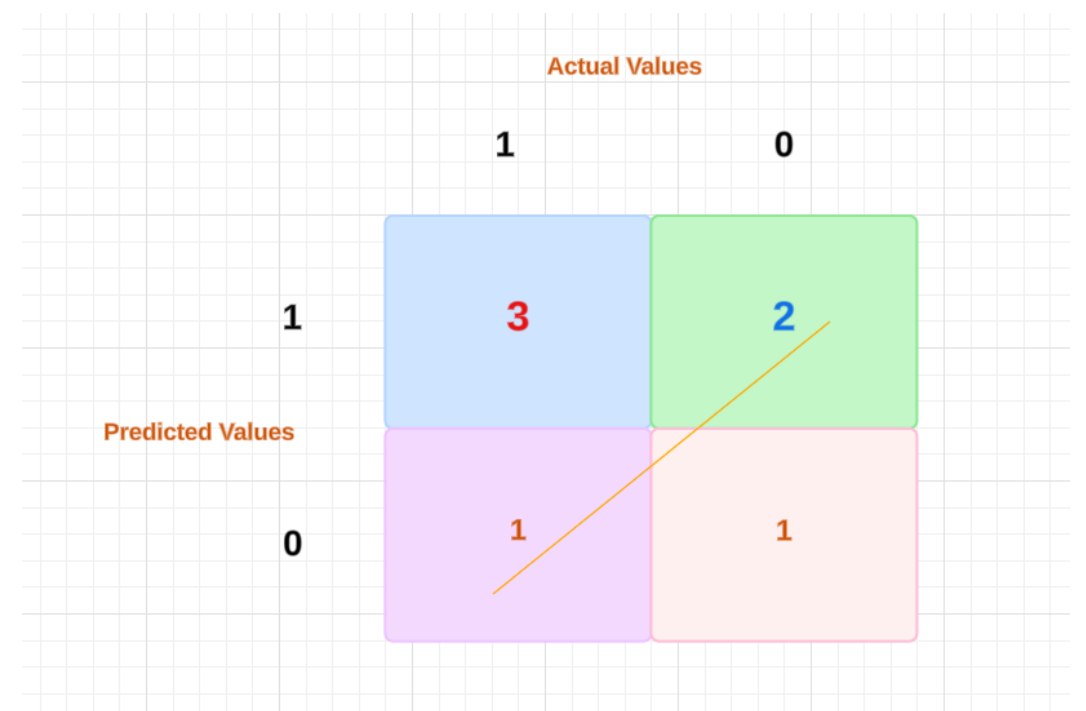
	Actual Values	
	1	0
1	3	2
Predicted Values		
0	1	1

X1	X2	Y	$\hat{y}$
A	Z	0	1
B	c	1	1
k	h	0	0
u	l	1	1
f	y	1	1
i	o	0	1
g	z	1	0

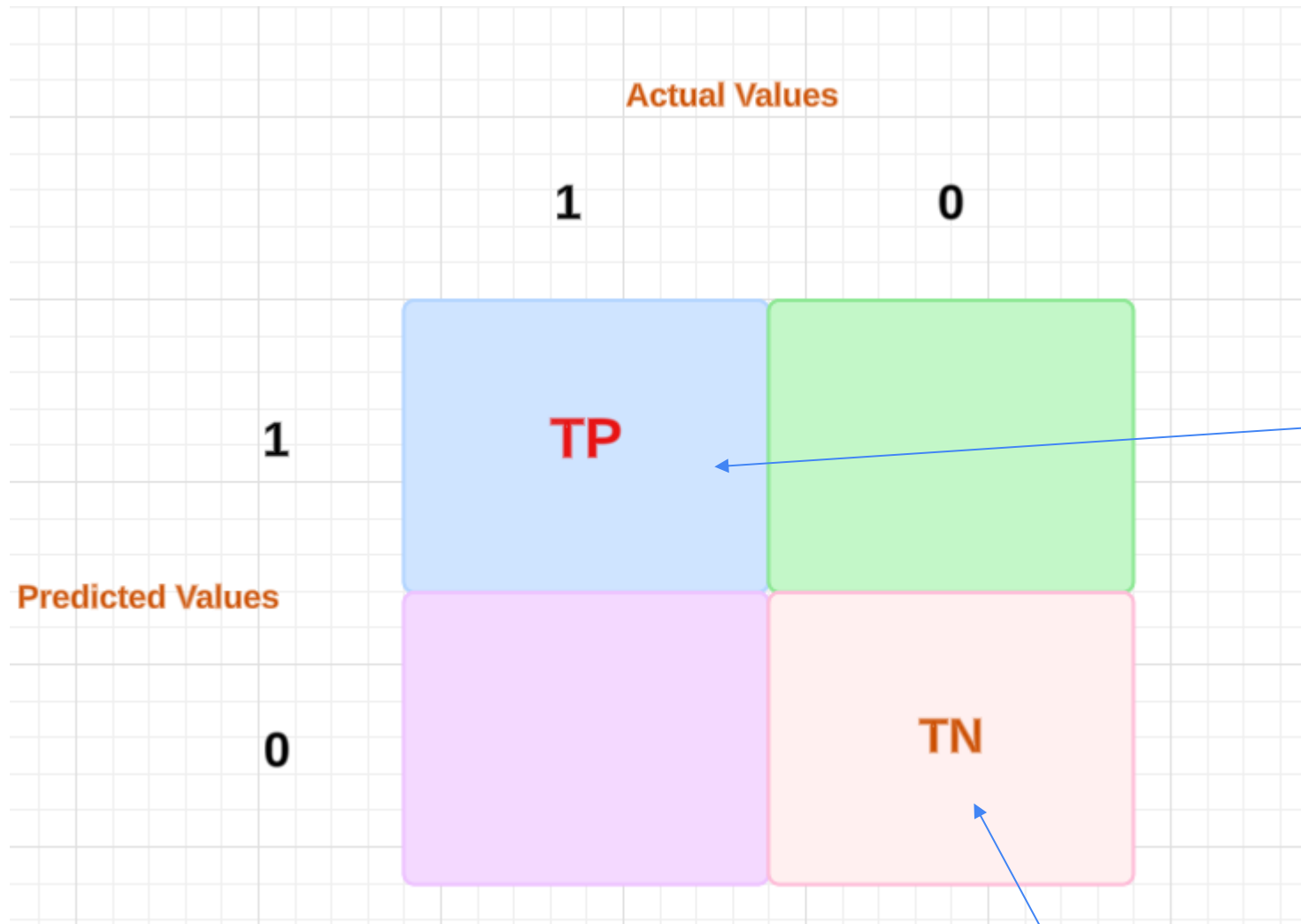
3 times it predicted **1** value correctly and **1 time** it predicted **0** value correctly



This Diagonal shows the **correct results**

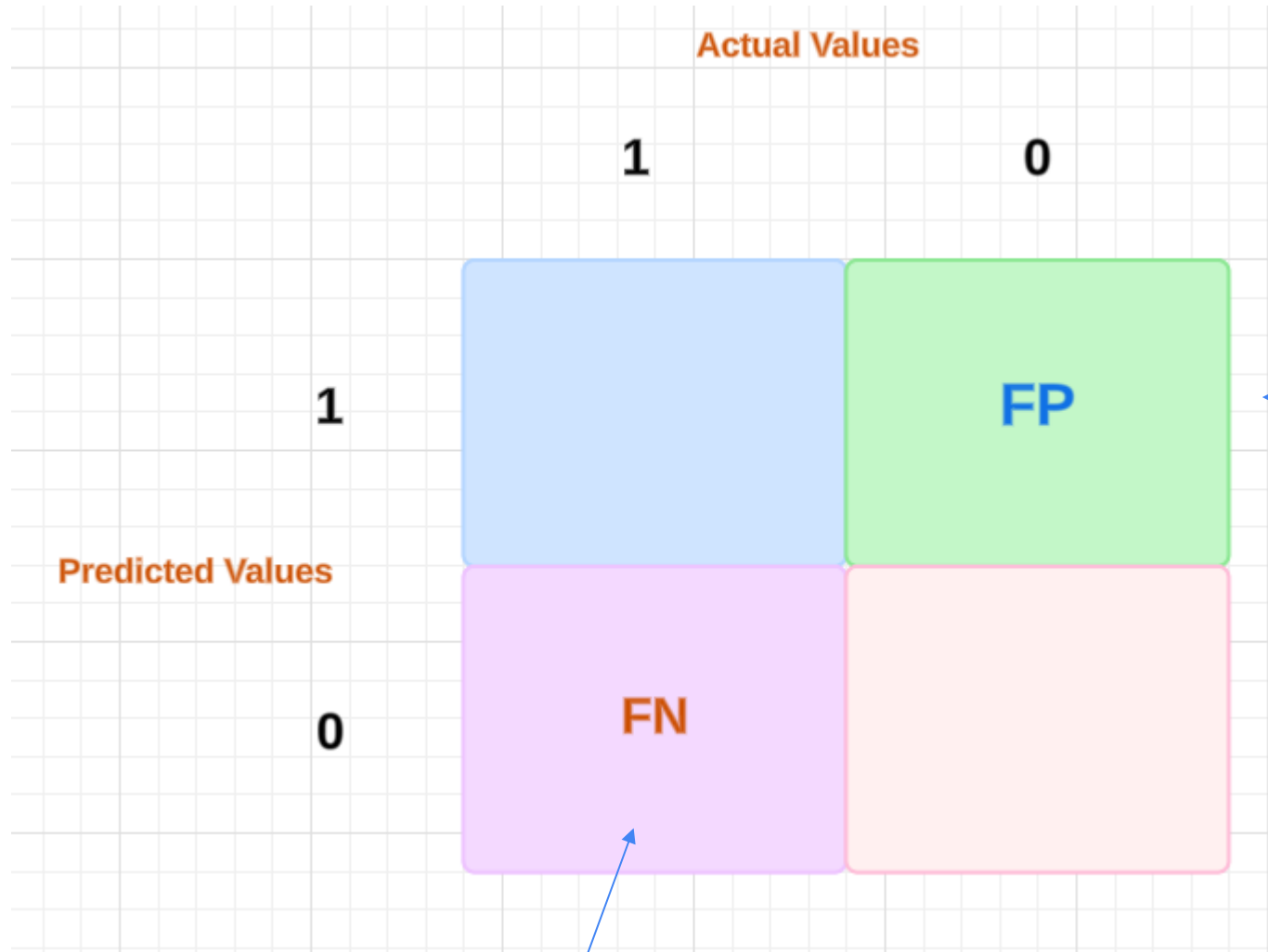


This Diagonal shows the **In correct results**



When My actual value is **1**
And Predicted value is also **1**
Then it is called **True Positive**

When My actual value is **0**
And Predicted value is also **0**
Then it is called **True Negative**



When My actual value is **0**
And Predicted value is **1**
Then it is called **False Positive**

When My actual value is **1**
And Predicted value is **0**
Then it is called **False Negative**

		Actual Values	
		1	0
Predicted Values	1	TP	FP
	0	FN	TN

So **Confusion Matrix** is a **table** that **visualizes** the performance of the **classification model** by showing the counts of **true positives, true negatives, false positives, and false negatives**.

Accuracy

The percentage of **correct predictions** made by the model (both true positives and true negatives).

		Actual Values	
		1	0
Predicted Values	1	TP	FP
	0	FN	TN

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

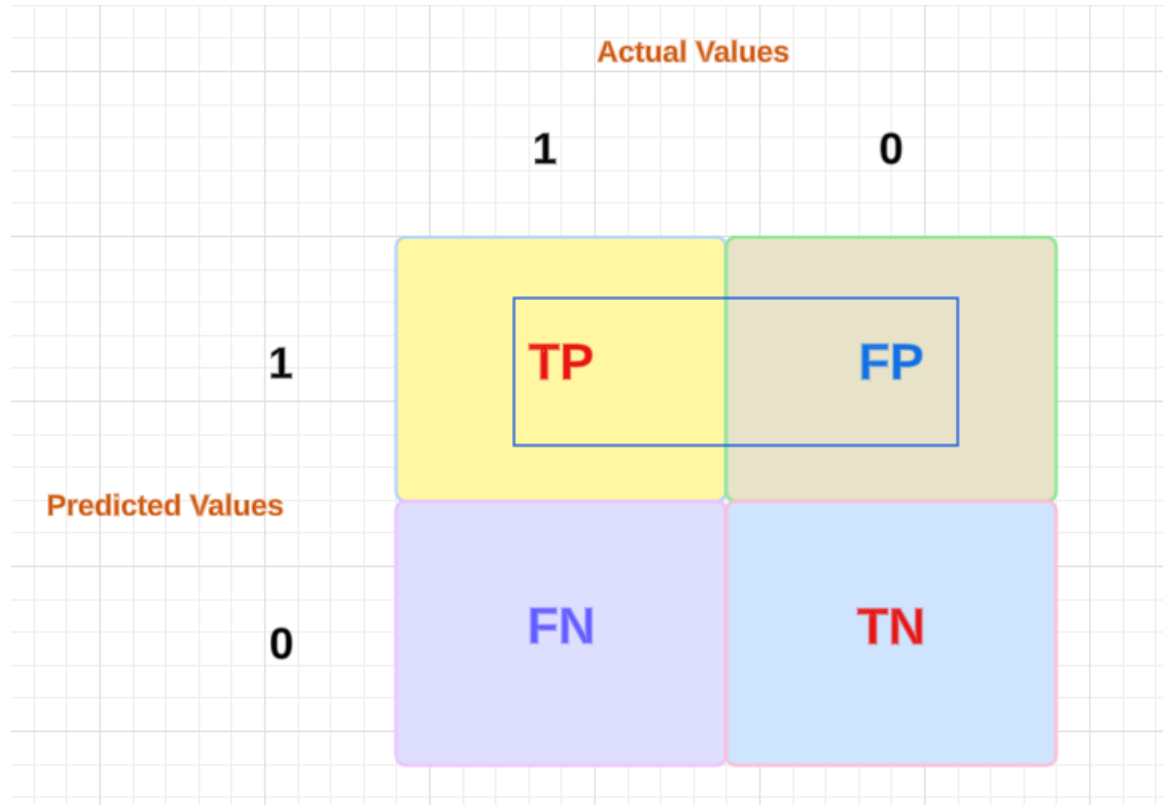
		Actual Values	
		1	0
Predicted Values	1	3	2
	0	1	1

$$\text{Accuracy} = \frac{3+1}{3+2+1+1}$$

$$\text{Accuracy} = \frac{4}{7}$$

Precision

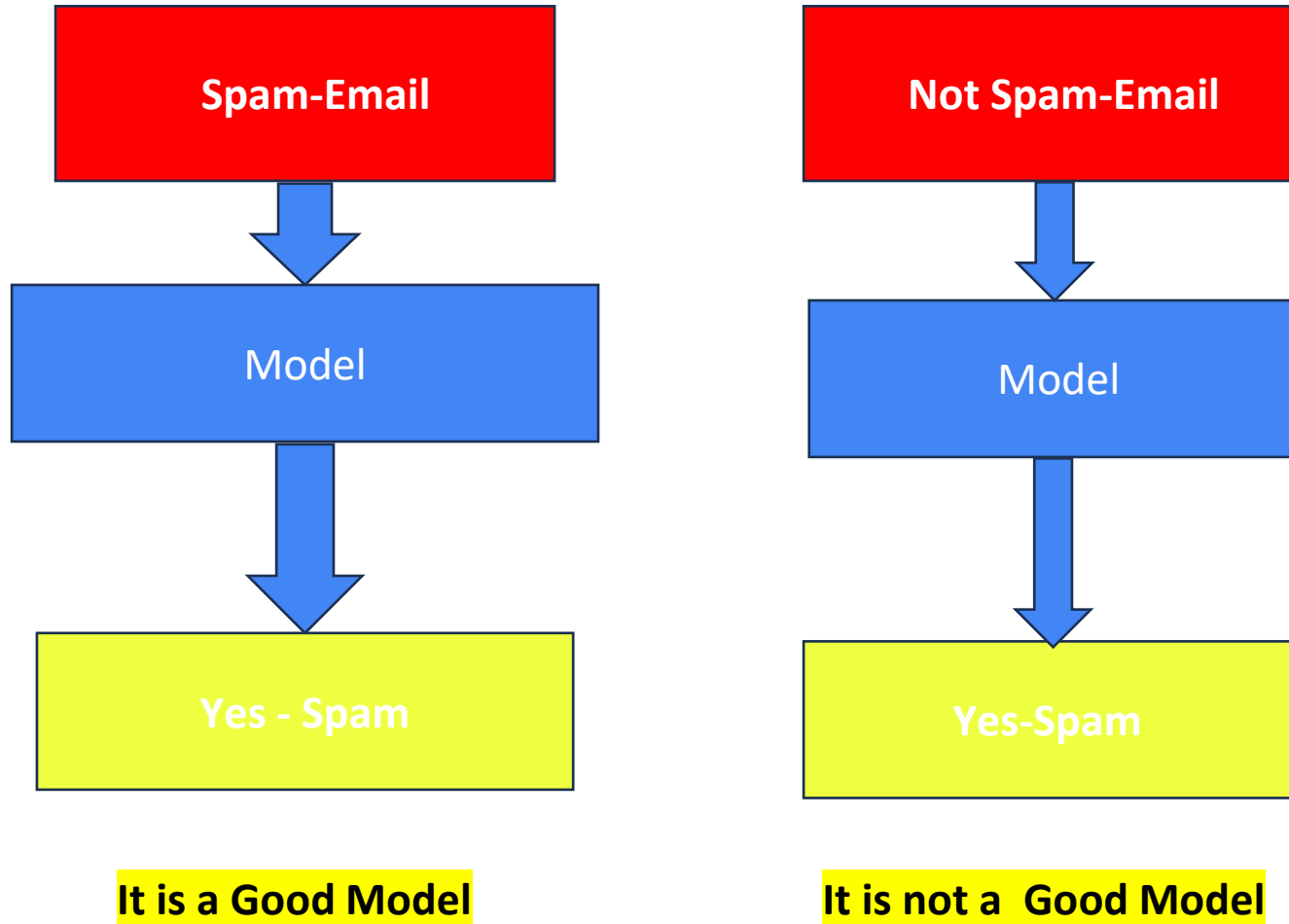
The **proportion** of **true positive predictions** out of **all positive predictions** made by the model.



$$\text{Precision} = \frac{TP}{TP + FP}$$

Use cases

1. Spam Email Classification:



		Actual Values	
		Spam 1	No Spam 0
Predicted Values	Spam 1	TP	FP
	No Spam 0	FN	TN

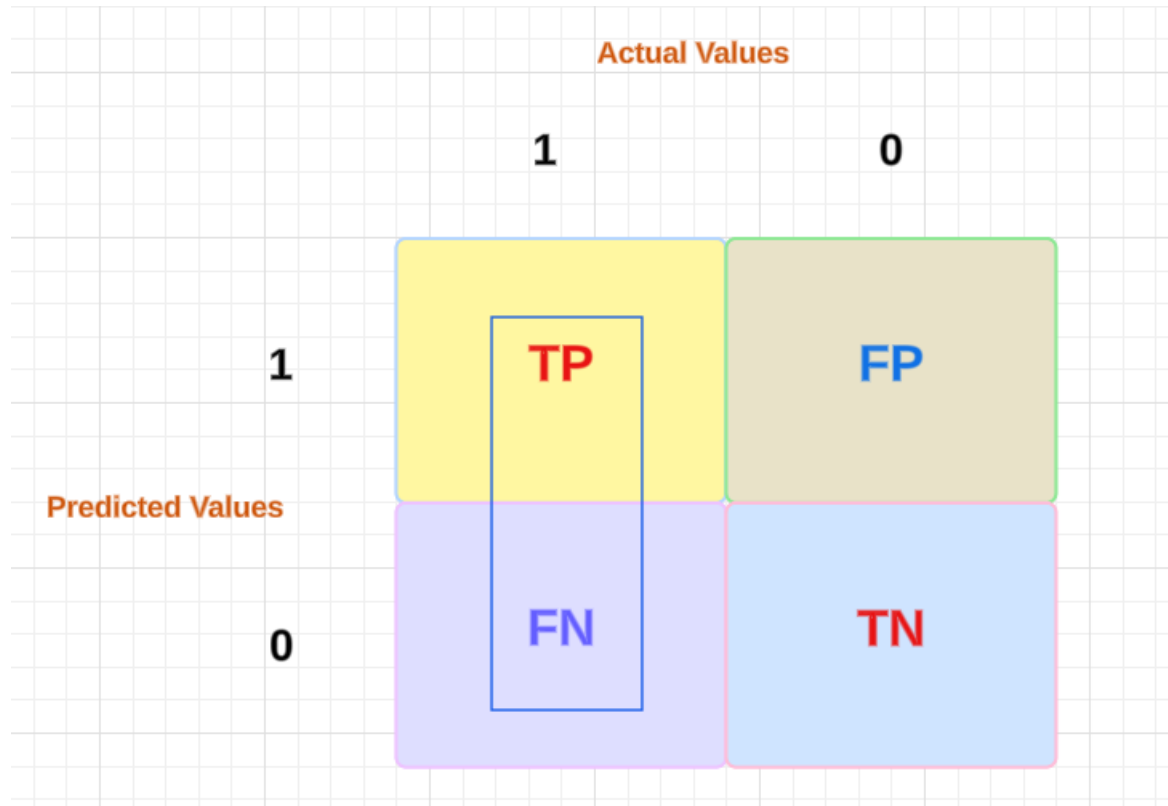
Mail is **Not –Spam** but my **Model** is **predicting** it as **Spam**.

This is called **False Positive**

In this case we will use **Precision** to **reduce** the **False positive**.

Recall

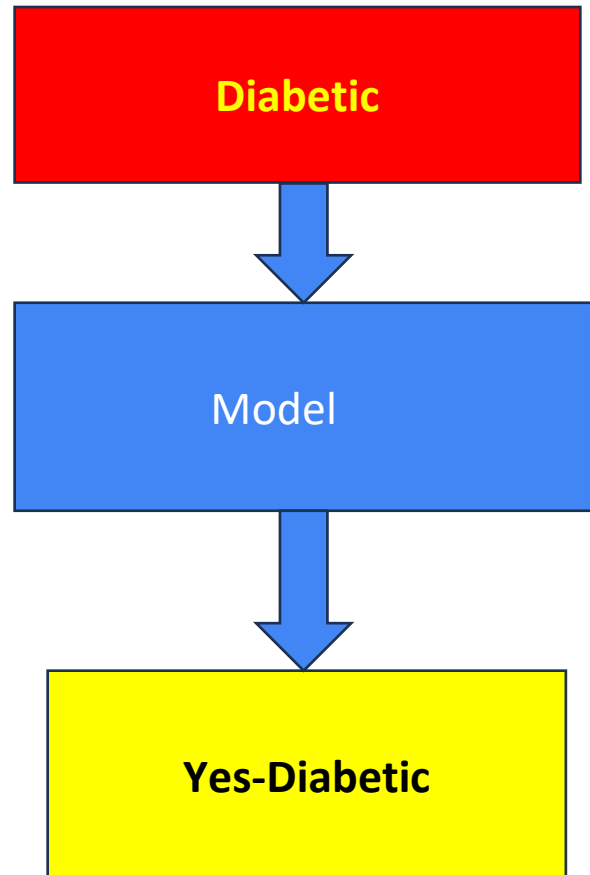
The proportion of **true positives prediction** out of all **actual positives**.



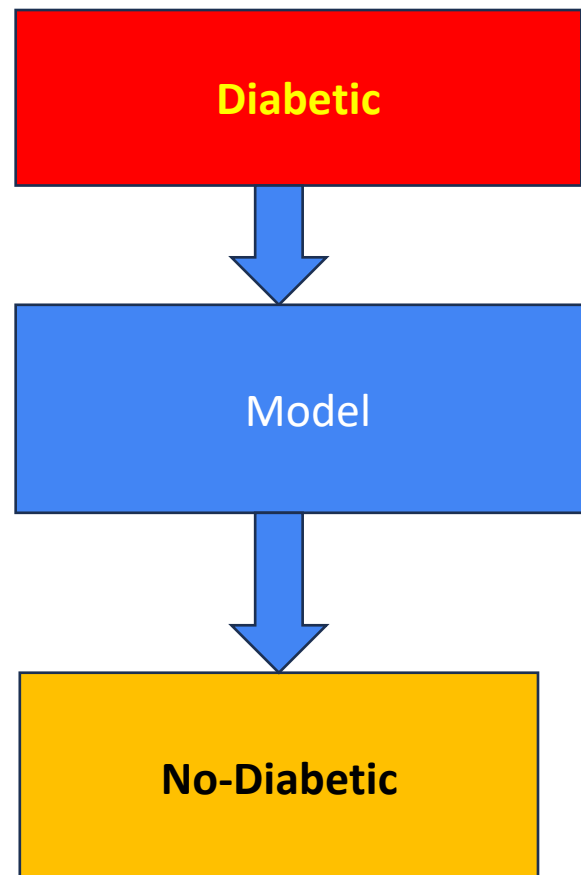
$$\text{Recall} = \frac{TP}{TP + FN}$$

Use Case

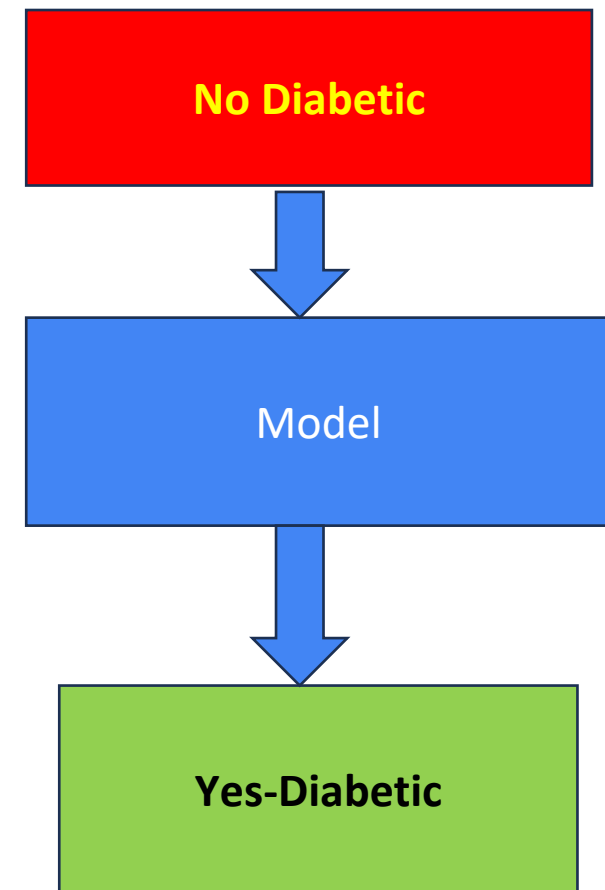
2. Diabetic Prediction



Good Model

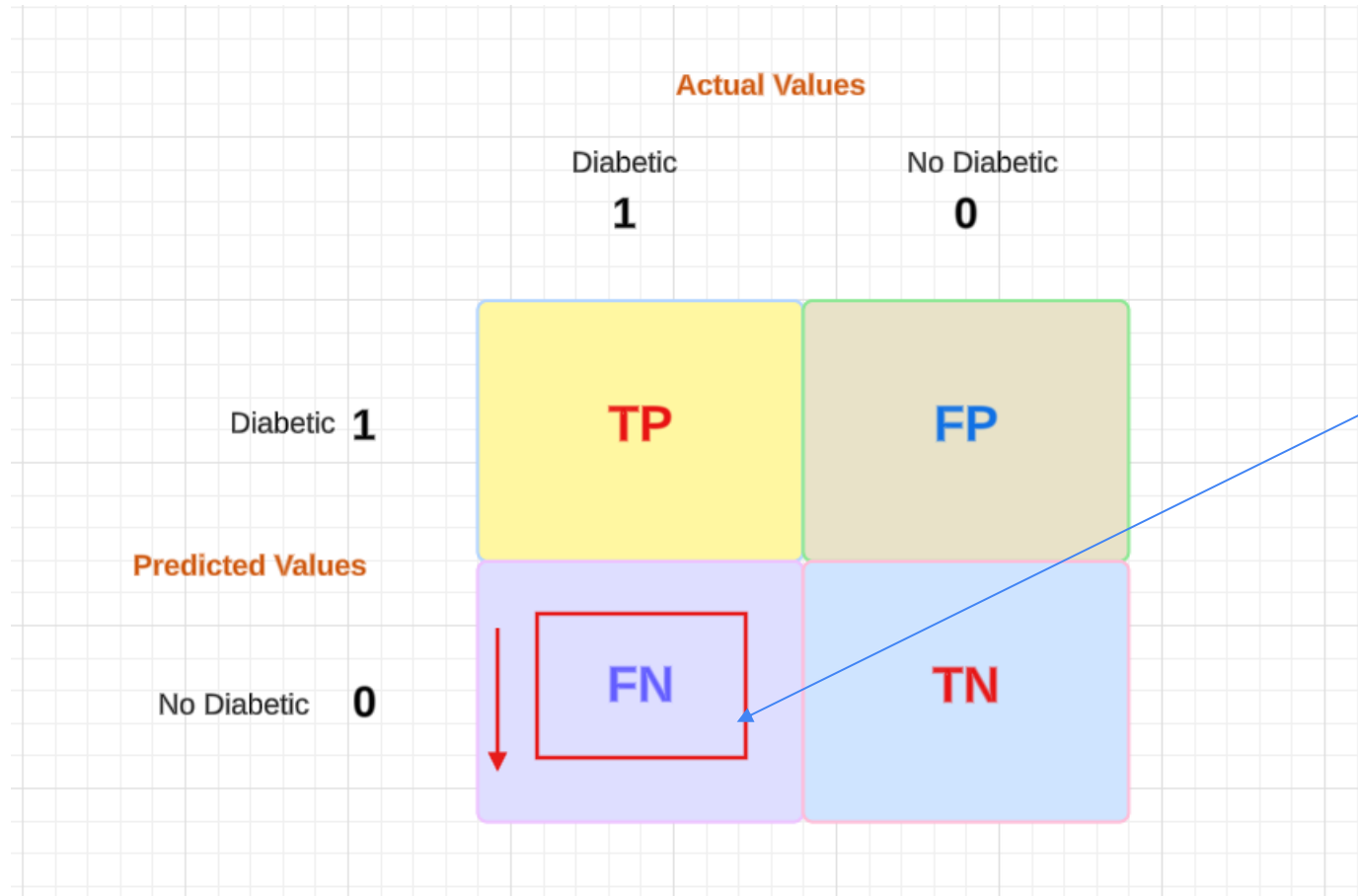


Bad Model



Not a good Model

Person may go for a second opinion



Person having **diabetic**
But our model predicting as **No diabetic**
This is called **False Negative**

In this case we will use **Recall** to **reduce the False Negative**.

F-Beta Score or F1-Score

● The **harmonic mean** of **precision** and **recall**. The F1 score provides a balance between precision and recall.

$$F\beta \text{ score} = (1 + \beta^2) \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is especially helpful when you need to find a **balance between precision and recall**, particularly in **imbalanced datasets**.

.....

1. If **FP & FN** are both important and we need to **reduce both of them**.

$$\beta = 1$$

$$\text{F1 score} = (1 + 1^2) \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

2. If **FP** is more important and we need to **reduce FP**

$$\beta = 0.5$$

$$\text{F0.5 score} = (1 + 0.5^2) \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= (1 + 0.25) \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

.....

3. If **FN** is more important and we need to **reduce FN**

$$\beta = 2$$

$$\text{F2 score} = (1 + 2^2) \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 5 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Question ?

Tomorrow the **stock market** will **crash or Not** ?

What you will Focus more ?

Reducing False positive

or

Reducing False Negative

.....

FP- Model predicts a market crash, but it does not actually happen.

Consequences- If you react to a false positive by **selling off assets**, you may **miss out on gains** if the market **doesn't crash** and *continues to rise*.

FN- Model fails to predict a crash, but the market does crash.

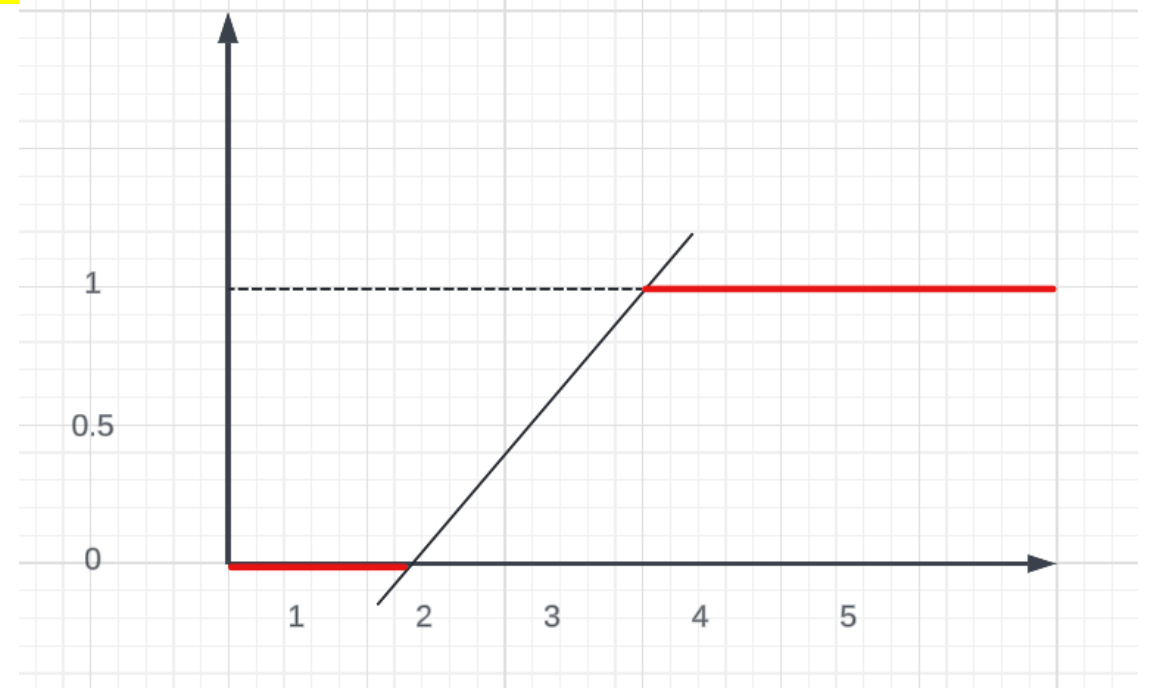
Consequences- The most severe consequence of a false negative is a **large financial loss** due to being **unprepared for the crash**.

■ ■ ■ ■ ■ ■ ■ ■

- For predicting stock market crashes, the primary focus is generally on **reducing false negatives** to ensure that you **don't** miss a real crash.
- **Missing a crash** can lead to much more significant **financial consequences** than taking precautionary measures based on a false alarm.
- **Most financial risk models** are designed to **avoid large, unexpected losses (crashes)**, so they tend to focus on **minimizing false negatives**.

Why we cannot use Linear regression?

- If I have an **outlier** my **best fit line** completely changes
- output also iam getting **>1 and <0**
- **To prevent the output >1 and <0** We have to **squash** the result
- This squash cannot be done in **Linear regression**



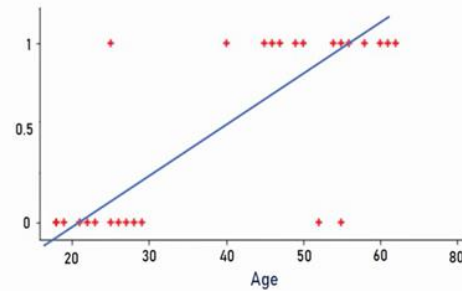
Sigmoid Activation Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

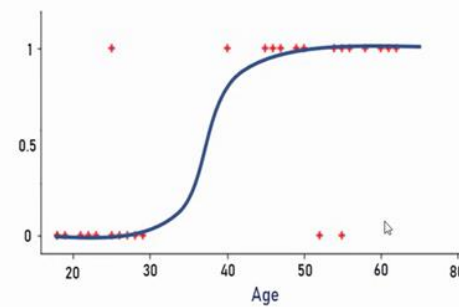
z is the input variable

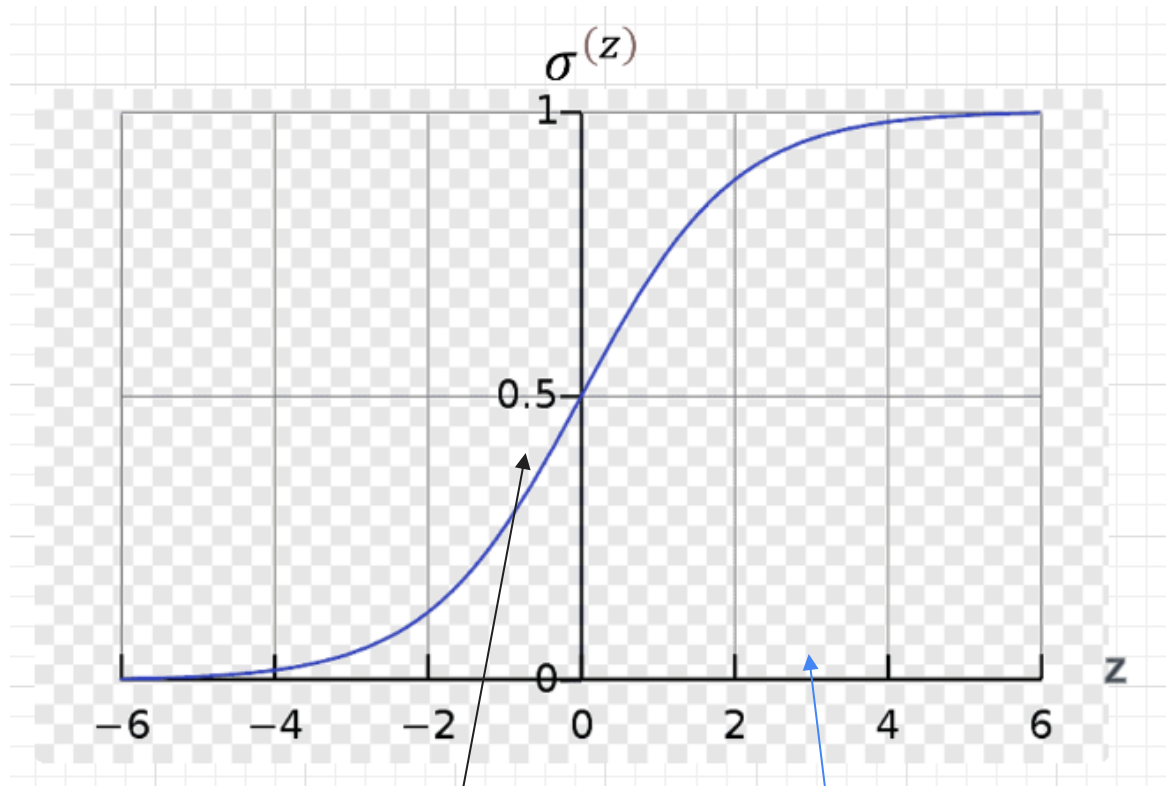
e is Euler's number (approximately 2.71828)

$$y = m * x + b$$



$$y = \frac{1}{1 + e^{-(m*x+b)}}$$

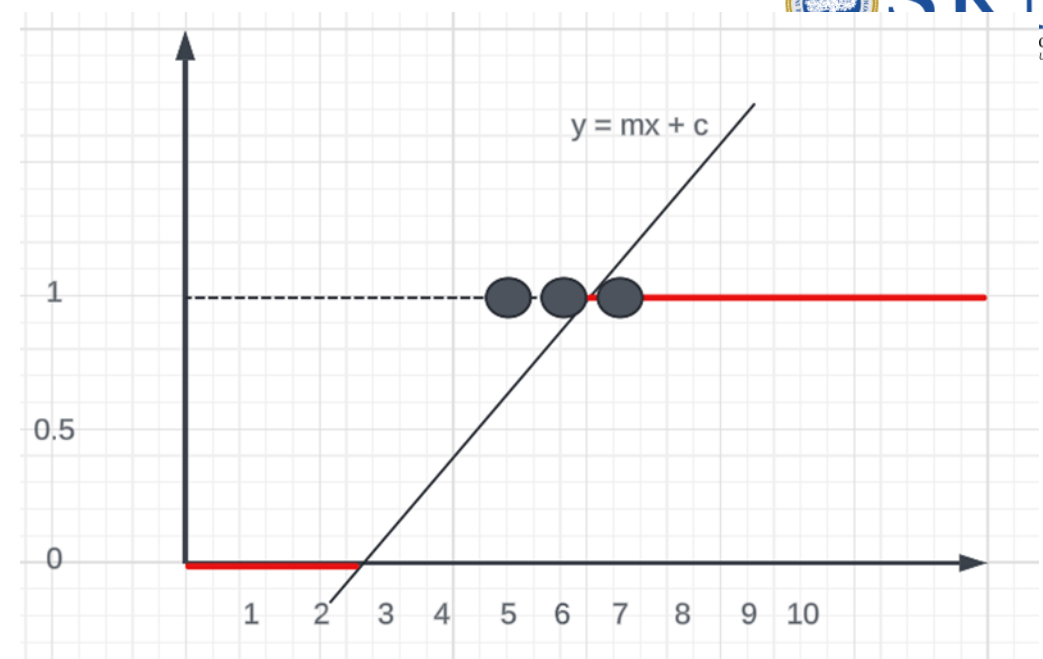




Sigmoid Curve

For any z value $\sigma(z)$ will be always between **0 to 1**

0.5 is the **Threshold value** by default used to classify the output



Linear Regression

$\sigma^{(z)}$ predicts the probability of the **positive class**
(typically **class 1**).

If $\sigma^{(z)} > 0.5$ the model predicts the **positive class.1**

If $\sigma^{(z)} < 0.5$ the model predicts the **Negative class.1**

However, you can adjust this **threshold** to increase **Recall**, depending on the problem's needs.

This is where the **ROC curve** comes in, showing how the **model's performance** changes as the **threshold shifts**.

ROC Curves

- The **ROC curve** (Receiver Operating Characteristic curve) serves to evaluate the performance of a **binary classification model** by showing how well **the model distinguishes** between **two classes**, typically **positive and negative**, across **different threshold settings**.

True Positive Rate (TPR) or Sensitivity or Recall

The proportion of **actual positives** that are correctly identified as Positive by the model .

$$TPR = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

False Positive Rate (FPR)

The proportion of **actual negatives** that are **incorrectly identified as positives** by the model.

$$FPR = \frac{\text{False Positives (FP)}}{\text{False Positives (FP)} + \text{True Negatives (TN)}}$$

Key Purposes of the ROC Curve

1. **Evaluating Classifier Performance Across Thresholds:**

The ROC curve shows how the **performance** of a **classifier** **changes** as the **decision threshold** is varied.

2. **Comparing Models:** ROC curves allow you to **compare the performance of different models** visually.